MicroVerse — Organism guide

30 organisms · 20 functions · generated 2026-10-27
All organisms are biosafety level 1. Functions marked 'from traits' are read directly from the numbers the simulation uses. Traits are textbook estimates and are not yet validated.

Part 1 — Functions: what organisms do

☀ Primary producer
Makes new organic matter from light and CO_2. Everything else in a food web ultimately lives on it.
In the simulation: Grows on light × nitrogen; needs no organic food.
Who: Synechococcus elongatus, Chlorella vulgaris, Scenedesmus obliquus, Navicula sp., Euglena gracilis

O_2 Oxygen producer
Releases oxygen while photosynthesising, which aerobic organisms then breathe.
In the simulation: Adds 1.4 mg O_2 per mg of new biomass to the dissolved-oxygen pool.
Who: Synechococcus elongatus, Chlorella vulgaris, Scenedesmus obliquus, Navicula sp., Euglena gracilis

🥛 Lactic acid fermenter
Turns sugar into lactic acid. This acidifies and preserves food (yoghurt, cheese, pickles).
In the simulation: A fixed fraction of the sugar it eats becomes lactate, which lowers the pH.
Who: Streptococcus thermophilus, Lactobacillus delbrueckii subsp. bulgaricus, Lactococcus lactis, Lactiplantibacillus plantarum, Leuconostoc mesenteroides

🍷 Alcohol fermenter
Turns sugar into ethanol and CO_2 (bread, beer, wine).
In the simulation: Releases ethanol into the shared pool, where others may eat it.
Who: Leuconostoc mesenteroides, Saccharomyces cerevisiae

pH↓ Acidifier
Lowers the pH of its surroundings. This can shut down acid-sensitive neighbours — and eventually itself.
In the simulation: Its lactate/acetate enters a charge balance that sets the pH everyone experiences.
Who: Streptococcus thermophilus, Lactobacillus delbrueckii subsp. bulgaricus, Lactococcus lactis, Lactiplantibacillus plantarum, Leuconostoc mesenteroides, Acetobacter aceti, Escherichia coli K-12

✂ Decomposer (enzyme producer)
Secretes enzymes that cut polymers (starch, protein, cellulose) into small sugars — food that others can then use too.
In the simulation: Converts the polymer pool into the sugar pool outside the cell, for everyone.
Who: Lactobacillus delbrueckii subsp. bulgaricus, Aspergillus oryzae, Penicillium roqueforti, Rhizopus oligosporus, Bacillus subtilis

⇄ Cross-feeder
Lives on a waste product of another organism (lactate, acetate or ethanol).
In the simulation: Eats from a pool that other organisms fill with their products.
Who: Propionibacterium freudenreichii, Acetobacter aceti, Penicillium roqueforti, Escherichia coli K-12, Azotobacter vinelandii, Halobacterium salinarum

♻ Aerobic scavenger
With oxygen, eats almost any small organic molecule — including leftovers from fermenters.
In the simulation: Uses several organic pools at once, but only when oxygen is present.
Who: Pseudomonas putida

🦠→ Bacterial grazer
Eats bacteria and other very small cells. Controls bacterial numbers and recycles nutrients.
In the simulation: Holling type II grazing on prey smaller than its size limit.
Who: Tetrahymena thermophila, Paramecium caudatum, Amoeba proteus, Vorticella convallaria, Brachionus calyciflorus, Philodina roseola

🌿→ Algal grazer
Eats microalgae. Drives classic predator–prey cycles.
In the simulation: Grazes prey up to a larger size limit that includes algal cells.
Who: Paramecium caudatum, Amoeba proteus, Brachionus calyciflorus, Philodina roseola

🔺 Top predator
Large enough to eat other grazers, not only bacteria and algae.
In the simulation: Its prey size limit is large enough to include protists.
Who: Amoeba proteus

🛡 Grazing-resistant
Too large (colonies, filaments) or armoured to be eaten by most grazers.
In the simulation: Marked not edible, so grazers ignore it.
Who: Aspergillus oryzae, Penicillium roqueforti, Rhizopus oligosporus, Scenedesmus obliquus, Methanosarcina barkeri

N_2 Nitrogen fixer
Takes nitrogen gas from the air, so it does not depend on dissolved nitrogen; returns it to the system when it dies.
In the simulation: Needs no dissolved nitrogen to grow.
Who: Azotobacter vinelandii

NH_4→ Nitrifier
Gets energy by oxidising ammonia. Part of the nitrogen cycle; competes with algae for nitrogen.
In the simulation: Consumes ammonium and oxygen; builds biomass from CO_2.
Who: Nitrosomonas europaea

CH_4 Methanogen
Turns acetate into methane at the end of anaerobic food chains (biogas). Oxygen kills it.
In the simulation: Eats only acetate, and only without oxygen.
Who: Methanosarcina barkeri

☀+🍽 Mixotroph
Can photosynthesise and also absorb organic food.
In the simulation: Photoautotroph that also lists organic substrates.
Who: Euglena gracilis

⊘O_2 Oxygen-sensitive
Grows only (or best) without oxygen. Needs others to remove oxygen, or a sealed vessel.
In the simulation: Its growth is multiplied by an oxygen-inhibition term.
Who: Lactobacillus delbrueckii subsp. bulgaricus, Propionibacterium freudenreichii

🧀 Food starter culture
Used deliberately to make a food.
Who: Streptococcus thermophilus, Lactobacillus delbrueckii subsp. bulgaricus, Lactococcus lactis, Lactiplantibacillus plantarum, Leuconostoc mesenteroides, Propionibacterium freudenreichii, Acetobacter aceti, Saccharomyces cerevisiae, Kluyveromyces lactis, Aspergillus oryzae, Penicillium roqueforti, Rhizopus oligosporus

🔬 Model organism
A standard lab organism: much of biology was first discovered in it.
Who: Lactococcus lactis, Saccharomyces cerevisiae, Bacillus subtilis, Escherichia coli K-12, Pseudomonas putida, Synechococcus elongatus, Chlorella vulgaris, Tetrahymena thermophila, Paramecium caudatum

🔥 Extremophile
Thrives where most life cannot (very salty, hot, acidic or dry).
Who: Philodina roseola, Halobacterium salinarum

Part 2 — Organisms

Lactic acid bacterium

Streptococcus thermophilus — ستربتوكوكس ثيرموفيلس
🥛 Lactic acid fermenter · pH↓ Acidifier · 🧀 Food starter culture
What it does: Fast early acidifier in yoghurt
Eats: Sugars (glucose, lactose…) · **Produces:** Lactate · **Oxygen:** Facultative — grows with or without oxygen
Best at: 42 °C, pH 6.5 · **Size:** ≈0.7–0.9 µm cocci
Bacteria › Bacillota › Bacilli › Lactobacillales › Streptococcaceae › Streptococcus › Streptococcus thermophilus (NCBI 1308) · Yoghurt, mozzarella

Lactobacillus delbrueckii subsp. bulgaricus — لاكتوباسيلس دلبروكي تحت نوع بلغاريكس
🥛 Lactic acid fermenter · pH↓ Acidifier · ✂ Decomposer (enzyme producer) · ⊘O_2 Oxygen-sensitive · 🧀 Food starter culture
What it does: Proteolysis and late acidification
Eats: Sugars (glucose, lactose…), Polymers (starch, protein, cellulose) · **Produces:** Lactate · **Oxygen:** Anaerobe — oxygen stops or harms it

Best at: 45 °C, pH 5.8 · **Size:** ≈0.5–0.8 × 2–9 µm rods
Bacteria › Bacillota › Bacilli › Lactobacillales › Lactobacillaceae › Lactobacillus › Lactobacillus delbrueckii › subsp. bulgaricus (NCBI 1585) · Yoghurt, some cheeses

Lactococcus lactis — لاكتوكوكس لاكتس
🦷 Lactic acid fermenter · pH↓ Acidifier · 🧀 Food starter culture · 🔬 Model organism
What it does: Mesophilic acidifier — works at room temperature where yoghurt starters are slow
Eats: Sugars (glucose, lactose…) · **Produces:** Lactate · **Oxygen:** Facultative — grows with or without oxygen
Best at: 30 °C, pH 6.3 · **Size:** ≈0.5–1.5 µm ovoid cocci
Bacteria › Bacillota › Bacilli › Lactobacillales › Streptococcaceae › Lactococcus › Lactococcus lactis (NCBI 1358) · Cheddar, feta, buttermilk; model organism for LAB genetics

Lactiplantibacillus plantarum — لاكتيبلانتيباسيلس بلانتاروم
🦷 Lactic acid fermenter · pH↓ Acidifier · 🧀 Food starter culture
What it does: Finishes vegetable fermentations after less acid-tolerant species stop
Eats: Sugars (glucose, lactose…) · **Produces:** Lactate, Acetate · **Oxygen:** Facultative — grows with or without oxygen
Best at: 33 °C, pH 6 · **Size:** ≈0.9–1.2 × 3–8 µm rods
Bacteria › Bacillota › Bacilli › Lactobacillales › Lactobacillaceae › Lactiplantibacillus › Lactiplantibacillus plantarum (NCBI 1590) · Pickles, sauerkraut, silage, probiotics

Leuconostoc mesenteroides — ليوكونوستوك ميزنتيرويدس
🦷 Lactic acid fermenter · 🍷 Alcohol fermenter · pH↓ Acidifier · 🧀 Food starter culture
What it does: Starts vegetable fermentations and produces CO_2 and aroma compounds
Eats: Sugars (glucose, lactose…) · **Produces:** Lactate, Acetate, Ethanol · **Oxygen:** Facultative — grows with or without oxygen
Best at: 28 °C, pH 6.3 · **Size:** ≈0.5–0.7 × 0.7–1.2 µm
Bacteria › Bacillota › Bacilli › Lactobacillales › Lactobacillaceae › Leuconostoc › Leuconostoc mesenteroides (NCBI 1245) · Sauerkraut, kimchi, dextran production

Acetic acid bacterium

Acetobacter aceti — أسيتوباكتر أسيتي
pH↓ Acidifier · ⇄ Cross-feeder · 🧀 Food starter culture
What it does: CROSS-FEEDER on yeast ethanol — turns wine into vinegar
Eats: Ethanol, Sugars (glucose, lactose…) · **Produces:** Acetate · **Oxygen:** Strict aerobe — needs oxygen
Best at: 30 °C, pH 5.5 · **Size:** ≈0.6 × 1–2 µm
Bacteria › Pseudomonadota › Alphaproteobacteria › Acetobacterales › Acetobacteraceae › Acetobacter › Acetobacter aceti (NCBI 435) · Vinegar production

Actinomycete

Propionibacterium freudenreichii — بروبيونيباكتيريوم فرويدنرايشي
⇄ Cross-feeder · ⊘O_2 Oxygen-sensitive · 🧀 Food starter culture
What it does: CROSS-FEEDER: lives on another organism's waste product (lactate)
Eats: Lactate, Sugars (glucose, lactose…) · **Produces:** Acetate · **Oxygen:** Anaerobe — oxygen stops or harms it
Best at: 30 °C, pH 6.7 · **Size:** ≈0.5 × 1–3 µm
Bacteria › Actinomycetota › Actinomycetes › Propionibacteriales › Propionibacteriaceae › Propionibacterium › Propionibacterium freudenreichii (NCBI 1744) · Emmental cheese, vitamin B12 production

Endospore-forming bacterium

Bacillus subtilis — باسيلس سبتيلس
✂ Decomposer (enzyme producer) · 🔬 Model organism
What it does: Fast aerobic decomposer and strong competitor for sugars
Eats: Polymers (starch, protein, cellulose), Sugars (glucose, lactose…) · **Produces:** Acetate · **Oxygen:** Facultative — grows with or without oxygen
Best at: 37 °C, pH 7 · **Size:** ≈0.8 × 2–6 µm rods
Bacteria › Bacillota › Bacilli › Bacillales › Bacillaceae › Bacillus › Bacillus subtilis (NCBI 1423) · Enzymes, probiotics, natto; the Gram-positive model organism

Enteric bacterium

Escherichia coli K-12 — إشريشيا كولاي سلالة K-12
pH↓ Acidifier · ⇄ Cross-feeder · 🔬 Model organism
What it does: The fastest grower here — shows how a fast competitor can take over a rich medium
Eats: Sugars (glucose, lactose…), Acetate · **Produces:** Acetate, Ethanol, Lactate · **Oxygen:** Facultative — grows with or without oxygen
Best at: 37 °C, pH 7 · **Size:** ≈0.5 × 2 µm rods
Bacteria › Pseudomonadota › Gammaproteobacteria › Enterobacterales › Enterobacteriaceae › Escherichia › Escherichia coli › K-12 (NCBI 83333) · The workhorse of molecular biology

Soil bacterium

Pseudomonas putida — سودوموناس بوتيدا
♻ Aerobic scavenger · 🔬 Model organism
What it does: SCAVENGER: cleans up fermentation products when oxygen is present
Eats: Sugars (glucose, lactose…), Acetate, Ethanol, Lactate · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 30 °C, pH 7 · **Size:** ≈0.7 × 2–3 µm rods
Bacteria › Pseudomonadota › Gammaproteobacteria › Pseudomonadales › Pseudomonadaceae › Pseudomonas › Pseudomonas putida (NCBI 303) · Bioremediation, biocatalysis

Azotobacter vinelandii — أزوتوباكتر فينلاندي
⇄ Cross-feeder · N_2 Nitrogen fixer
What it does: NITROGEN SUPPLIER: keeps growing when everyone else runs out of nitrogen, and releases it when it dies
Eats: Sugars (glucose, lactose…), Acetate · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 30 °C, pH 7.2 · **Size:** ≈2–4 µm, unusually large cells
Bacteria › Pseudomonadota › Gammaproteobacteria › Pseudomonadales › Pseudomonadaceae › Azotobacter › Azotobacter vinelandii (NCBI 354) · Biofertiliser, alginate production, nitrogenase research

Chemolithotroph

Nitrosomonas europaea — نيتروزوموناس يوروبيا
NH_4→ Nitrifier
What it does: NITROGEN COMPETITOR: removes the ammonia that algae and heterotrophs need
Eats: Inorganic nitrogen · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 30 °C, pH 7.8 · **Size:** ≈0.8 × 1.5 µm
Bacteria › Pseudomonadota › Betaproteobacteria › Nitrosomonadales › Nitrosomonadaceae › Nitrosomonas › Nitrosomonas europaea (NCBI 915) · Wastewater treatment, aquarium cycling, nitrogen cycle teaching

Cyanobacterium

Synechococcus elongatus — سينيكوكوكس إيلونجاتس
☀ Primary producer · O_2 Oxygen producer · 🔬 Model organism
What it does: OXYGEN PRODUCER and primary producer, like the algae but prokaryotic
Eats: Inorganic nitrogen · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 38 °C, pH 8 · **Size:** ≈1 × 2–3 µm
Bacteria › Cyanobacteriota › Cyanophyceae › Synechococcales › Synechococcaceae › Synechococcus › Synechococcus elongatus (NCBI 32046) · Photosynthesis research, biotechnology chassis

Archaeon

Methanosarcina barkeri — ميثانوسوركينا باركري
CH_4 Methanogen · 🛡️ Grazing-resistant
What it does: FINAL STEP of anaerobic food chains: removes the acetate everyone else leaves behind
Eats: Acetate · **Produces:** — · **Oxygen:** Anaerobe — oxygen stops or harms it
Best at: 37 °C, pH 7 · **Size:** Cells ≈1.5–2 µm, in packets
Archaea › Methanobacteriota › Methanomicrobia › Methanosarcinales › Methanosarcinaceae › Methanosarcina › Methanosarcina barkeri (NCBI 2208) · Biogas production, carbon cycle research

Halobacterium salinarum — هالوباكتيريوم سالیناروم
⇄ Cross-feeder · 🔥 Extremophile
What it does: Extremophile example — salinity is not modelled yet, so treat its growth here as indicative only
Eats: Sugars (glucose, lactose…), Acetate · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 42 °C, pH 7.2 · **Size:** ≈0.5 × 4 µm
Archaea › Methanobacteriota › Halobacteria › Halobacteriales › Halobacteriaceae › Halobacterium › Halobacterium salinarum (NCBI 2242) · Bioenergetics, astrobiology, salt-tolerance research

Yeast

Saccharomyces cerevisiae — ساكارواميسيس سيريفيزي
🍷 Alcohol fermenter · 🧀 Food starter culture · 🔬 Model organism
What it does: Main fermenter; its ethanol feeds Acetobacter, its CO_2 raises dough
Eats: Sugars (glucose, lactose…) · **Produces:** Ethanol, Acetate · **Oxygen:** Facultative — grows with or without oxygen
Best at: 32 °C, pH 5 · **Size:** ≈3–8 µm
Eukaryota › Fungi › Ascomycota › Saccharomycetes › Saccharomycetales › Saccharomycetaceae › Saccharomyces › Saccharomyces cerevisiae (NCBI 4932) · Bread, beer, wine, bioethanol; the model eukaryote

Kluyveromyces lactis — كلويفيروميسيس لاكتس
🧀 Food starter culture
What it does: Turns whey sugar into biomass
Eats: Sugars (glucose, lactose…) · **Produces:** Ethanol · **Oxygen:** Strict aerobe — needs oxygen
Best at: 30 °C, pH 5.5 · **Size:** ≈3–6 µm
Eukaryota › Fungi › Ascomycota › Saccharomycetes › Saccharomycetales › Saccharomycetaceae › Kluyveromyces › Kluyveromyces lactis (NCBI 28985) · Lactase enzyme, kefir, whey valorisation

Filamentous fungus

Aspergillus oryzae — أسبرجيلس أوريزي
✂ Decomposer (enzyme producer) · 🛡️ Grazing-resistant · 🧀 Food starter culture
What it does: DECOMPOSER / opener: unlocks polymers that other organisms cannot eat
Eats: Polymers (starch, protein, cellulose), Sugars (glucose, lactose…) · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 32 °C, pH 5.5 · **Size:** Hyphae 3–10 µm wide, colonies in mm
Eukaryota › Fungi › Ascomycota › Eurotiomycetes › Eurotiales › Aspergillaceae › Aspergillus › Aspergillus oryzae (NCBI 5062) · Soy sauce, miso, sake, industrial enzymes

Penicillium roqueforti — بنيسيليوم روكفورتي
✂ Decomposer (enzyme producer) · ⇄ Cross-feeder · 🛡️ Grazing-resistant · 🧀 Food starter culture
What it does: Ripening mould — changes texture and flavour rather than acidity
Eats: Polymers (starch, protein, cellulose), Sugars (glucose, lactose…), Lactate · **Produces:** — · **Oxygen:** Microaerophile — needs a little oxygen
Best at: 22 °C, pH 5.5 · **Size:** Hyphae 2–5 µm wide
Eukaryota › Fungi › Ascomycota › Eurotiomycetes › Eurotiales › Aspergillaceae › Penicillium › Penicillium roqueforti (NCBI 5082) · Roquefort, Gorgonzola, Stilton

Rhizopus oligosporus — ريزوبس أوليغوسبورس
✂ Decomposer (enzyme producer) · 🛡️ Grazing-resistant · 🧀 Food starter culture
What it does: Decomposer that makes protein available
Eats: Polymers (starch, protein, cellulose), Sugars (glucose, lactose…) · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 37 °C, pH 5.5 · **Size:** Wide aseptate hyphae 5–15 µm
Eukaryota › Fungi › Mucoromycota › Mucoromycetes › Mucorales › Rhizopodaceae › Rhizopus › Rhizopus microsporus var. oligosporus (NCBI 64495) · Tempeh

Green alga

Chlorella vulgaris — كلوريلا فولغاريس
☀ Primary producer · O_2 Oxygen producer · 🔬 Model organism
What it does: PREY / primary producer: feeds the grazers and oxygenates the water
Eats: Inorganic nitrogen · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 28 °C, pH 7.5 · **Size:** 2–10 µm diameter
Eukaryota › Viridiplantae › Chlorophyta › Trebouxiophyceae › Chlorellales › Chlorellaceae › Chlorella › Chlorella vulgaris (NCBI 3077) · Food supplement, biofuel, wastewater research

Scenedesmus obliquus — سينيديسمس أوبليكوس
☀ Primary producer · O_2 Oxygen producer · 🛡️ Grazing-resistant
What it does: GRAZING-RESISTANT producer: too big for small grazers to swallow
Eats: Inorganic nitrogen · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 27 °C, pH 7.5 · **Size:** Cells 5–15 µm, colonies up to 60 µm
Eukaryota › Viridiplantae › Chlorophyta › Chlorophyceae › Sphaeropleales › Scenedesmaceae › Scenedesmus › Scenedesmus obliquus (NCBI 3088) · Wastewater treatment, biodiesel, inducible-defence research

Diatom

Navicula sp. — نافيكولا
☀ Primary producer · O_2 Oxygen producer
What it does: Cold-tolerant producer; a staple food for grazers
Eats: Inorganic nitrogen · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 22 °C, pH 7.8 · **Size:** ≈10–60 µm long
Eukaryota › Sar › Stramenopiles › Bacillariophyta › Bacillariophyceae › Naviculales › Naviculaceae › Navicula (NCBI 55459) · Water quality indicator, forensic diatom tests

Flagellate protist

Euglena gracilis — يوغلينا غراسيلس
☀ Primary producer · O_2 Oxygen producer · ☀+🍽️ Mixotroph
What it does: Survives both as a producer and as a consumer — hard to out-compete
Eats: Inorganic nitrogen, Acetate, Sugars (glucose, lactose…) · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 27 °C, pH 6.5 · **Size:** ≈35–60 µm long
Eukaryota › Discoba › Euglenozoa › Euglenida › Euglenaceae › Euglena › Euglena gracilis (NCBI 3039) · Photobiology teaching, nutrition supplements

Ciliate protist

Tetrahymena thermophila — تيتراهيمينا ثيرموفيلا

🦠 → Bacterial grazer · 🔬 Model organism
What it does: BACTERIAL GRAZER: controls bacterial numbers and recycles nitrogen
Eats: other organisms · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 35 °C, pH 7 · **Size:** ≈30–50 µm long
Eukaryota › Sar › Alveolata › Ciliophora › Oligohymenophorea › Hymenostomatida › Tetrahymenidae › Tetrahymena thermophila (NCBI 5911) · Model organism (telomeres, ribozymes — two Nobel-prize subjects)

Paramecium caudatum — باramسيوم كوداتوم
🦠 → Bacterial grazer · 🌿 → Algal grazer · 🔬 Model organism
What it does: Large grazer — the classic Gause competition experiment organism
Eats: other organisms · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 25 °C, pH 7 · **Size:** ≈150–300 µm long
Eukaryota › Sar › Alveolata › Ciliophora › Oligohymenophorea › Peniculida › Parameciidae › Paramecium caudatum (NCBI 5885) · Teaching, competition and predation experiments

Sessile ciliate protist

Vorticella convallaria — فورتيسيلا
🦠 → Bacterial grazer
What it does: Attached filter feeder — an indicator of healthy wastewater sludge
Eats: other organisms · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 24 °C, pH 7.2 · **Size:** Bell ≈40–80 µm on a contractile stalk up to 200 µm
Eukaryota › Sar › Alveolata › Ciliophora › Oligohymenophorea › Sessilida › Vorticellidae › Vorticella (NCBI 5974) · Wastewater treatment indicator

Amoeboid protist

Amoeba proteus — أميبا بروتيوس
🦠 → Bacterial grazer · 🌿 → Algal grazer · 🔺 Top predator
What it does: TOP PREDATOR: eats the grazers as well as the bacteria
Eats: other organisms · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 22 °C, pH 7 · **Size:** ≈250–750 µm
Eukaryota › Amoebozoa › Tubulinea › Euamoebida › Amoebidae › Amoeba › Amoeba proteus (NCBI 5775) · Classic microscopy teaching organism

Rotifer — microscopic multicellular animal

Brachionus calyciflorus — براكيونس كاليسيفلورس
🦠 → Bacterial grazer · 🌿 → Algal grazer
What it does: PREDATOR / grazer on algae
Eats: other organisms · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 25 °C, pH 7.2 · **Size:** ≈150–350 µm long
Eukaryota › Metazoa › Rotifera › Monogononta › Ploima › Brachionidae › Brachionus › Brachionus calyciflorus (NCBI 104777) · Ecotoxicology tests, live feed in aquaculture

Philodina roseola — فيلودينا
🦠 → Bacterial grazer · 🌿 → Algal grazer · 🔥 Extremophile
What it does: Detritus-and-bacteria feeder that tolerates harsh conditions
Eats: other organisms · **Produces:** — · **Oxygen:** Strict aerobe — needs oxygen
Best at: 22 °C, pH 7 · **Size:** ≈300–500 µm
Eukaryota › Metazoa › Rotifera › Bdelloidea › Philodinida › Philodinidae › Philodina › Philodina roseola (NCBI 96728) · Research on desiccation survival and asexual evolution